Proposition 1 is now a corollary of Proposition 4, which follows by setting $\alpha = e^{-\frac{t^2 mn}{2K(m+n)}}$ and solving for t to obtain a test of level α.

A.2 Proof of Proposition 3

To find $n_{\mathrm{XY}_l^l}$, we use our implementation of MMDEW and the On-Line Encyclopedia of Integer Sequences (OEIS) to discover that $n_{\mathrm{XY}_l^l}$ follows the sequence $1, 2, 8, 24, 64, 160, \ldots$ for $l = 0, 1, 2, \ldots$. Thus

$$n_{\mathrm{XY}_l^l} = 2^l l, \quad \text{for } l > 0 \tag{10}$$

and $n_{\mathrm{XY}_0^0} = 1$ (Sloane, 1999b).

To find n_{XX_l}, notice that n_{XX_l} only changes when one merges two windows, which happens for windows of the same size $n_{\mathrm{XX}_{l-1}}$. The algorithm adds to this $2 \cdot n_{\mathrm{XY}_{l-1}^{l-1}}$ terms, see (7), and, for $l = 0, 1, 2, \ldots$, we obtain the recurrence relation

$$n_{\mathrm{XX}_l} = \begin{cases} 1 & \text{if } l = 0, \\ 4 & \text{if } l = 1, \\ 2 \cdot n_{\mathrm{XX}_{l-1}} + 2 \cdot n_{\mathrm{XY}_{l-1}^{l-1}} & \text{if } l > 1, \end{cases}$$

with $n_{\mathrm{XX}_{-1}} := 0$. Now write

$$n_{\mathrm{XX}_l} = 2 \cdot n_{\mathrm{XX}_{l-1}} + l \cdot 2^l - 2^l + 2 \cdot [l = 0] + 2 \cdot [l = 1], \tag{11}$$

where the brackets are equal to one if their argument is true and zero otherwise (using Iverson's convention; Graham et al. 1994). To find a closed-form expression for (11), we define the ordinary generating function $A(z) = \sum_l a_l z^l$. Now, we multiply (11) by z_l and sum on l, to obtain

$$A(z) = \frac{-8z^3 + 2z - 1}{(2z-1)^3}$$

after some algebra, so that

$$n_{\mathrm{XX}_l} = [z^l] \frac{-8z^3 + 2z - 1}{(2z-1)^3},$$

where $[z^l]$ is the coefficient of z^l in the series expansion of the generating function $A(z)$. To extract coefficients, we first decompose $A(z)$ as

$$A(z) = \frac{3}{1-2z} - \frac{2}{(1-2z)^2} + \frac{1}{(1-2z)^3} - 1,$$

which allows us to then find the coefficients as

$$[z^l] \frac{3}{1-2z} \overset{(a)}{=} 3 \cdot 2^l, \qquad [z^l] - \frac{2}{(2z-1)^2} \overset{(b)}{=} -(l+1)2^{l+1}, \qquad [z^l] \frac{1}{(1-2z)^3} \overset{(c)}{=} (l+1)(l+2)2^{l-1},$$

where Graham et al. (1994, Table 335) implies (a), (b) is (10) shifted, and (c) is Sloane (1999a) shifted. We omit the last term as it corresponds to $[z^0]$, which we do not need. Now, adding all terms gives

$$3 \cdot 2^l - (l+1)2^{l+1} + (l+1)(l+2)2^{l-1} = 2^{l-1}(l^2 - l + 4),$$

concluding the proof.

B External results

To proof Proposition 1, we recall McDiarmid's concentration inequality (Vershynin, 2018).

Theorem 1 (Bounded differences inequality)**.** *Let $X = (X_1, \dots, X_n)$ be a random vector with independent components. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a measurable function. Assume that the value of $f(x)$ can change by at most $c_i > 0$ under an arbitrary change of a single coordinate of $x = (c_1, \dots, c_n) \in \mathbb{R}^n$. Then, for any $t > 0$, we have*

$$P\{f(X) - \mathbb{E}f(X) \geq t\} \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

C Hyperparameter optimization settings

We collect the hyperparameter choices that we tested in our experiments on real-world classification data (Section 5.2) in Table 3 and refer to the respective original publications for additional information on the parameter settings.

Table 3: Values chosen for the parameter optimization.

Algorithm	Parameters	Parameter values
MMDEW	α	$\alpha \in \{0.001, 0.01, 0.1, 0.2\}$
ADWINK	δ, k	$\delta \in \{0.05, 0.1, 0.2, 0.9, 0.99\}$, $k \in \{0.01, 0.02, 0.05, 0.1, 0.2\}$
D3	$\omega, \rho, \tau, \mathrm{d}$	$\omega \in \{100, 200, 500\}$, $\rho \in \{0.1, 0.3, 0.5\}$, $\tau \in \{0.7, 0.8, 0.9\}$, $\mathrm{d} = 1$
IBDD	m, w	$m \in \{10, 20, 50, 100\}$, $w \in \{20, 100, 200, 300\}$
NEWMA	ω, α	$\omega \in \{20, 50, 100\}$, $\alpha \in \{0.01, 0.02, 0.05, 0.1\}$
Scan B	B, ω, α	$B \in \{2, 3\}$, $\omega \in \{100, 200, 300\}$, $\alpha \in \{0.01, 0.05\}$
WATCH	$\epsilon, \kappa, \mu, \omega$	$\epsilon \in \{1, 2, 3\}$, $\kappa \in \{25, 50, 100\}$, $\mu \in \{10, 20, 50, 100, 1000, 2000\}$, $\omega \in \{100, 250, 500, 1000\}$

D Additional experiments

In this section, we collect additional experiments on synthetic data. In Appendix D.1, we align the ARLs of the kernel-based approaches and compare their respective EDD/MTD. In Appendix D.2, we show that our MMD-based approach detects changes in the covariance structure of multivariate data, which aggregated univariate approaches cannot detect reliably.

D.1 EDD/MTD of kernel-based approaches on synthetic data

The following experiments compare the EDD of the kernel-based online change detection approaches considered in the main text and the related work for a fixed ARL on toy data, extending the experiments of Wei & Xie (2022, Figure 4). We note that all approaches considered in this section compute, for each new observation, a test statistic and compare the statistic to a threshold. If the statistic exceeds the threshold, a change is flagged. In the case of the proposed algorithm, multiple test statistics (one for each possible split) are computed. To allow for a comparison, we select the maximum MMD value across all splits, that is, for MMD, we consider the stopping rule

$$T'' = \inf\left\{t : \max_{s=1,\dots,l} \mathrm{MMD}\left(\bigcup_{i=s+1}^{l} \mathrm{X}_i, \bigcup_{i=0}^{s} \mathrm{X}_i\right) \geq b\right\}, \tag{12}$$

with a fixed $b > 0$, instead of (9) as done in Section 5. Similarly, for MMDEW, we consider (12) but with subsampling applied to the X_i-s. The experimental setup is as follows.

To achieve a fixed target ARL $\mathbb{E}_{H_0}T$ for a given stopping time T, we run 25 Monte Carlo simulations on 150,000 samples from $\mathbb{P} = \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ with $d = 20$ and select b as the $1 - 1/(\text{target ARL})$-quantile of the

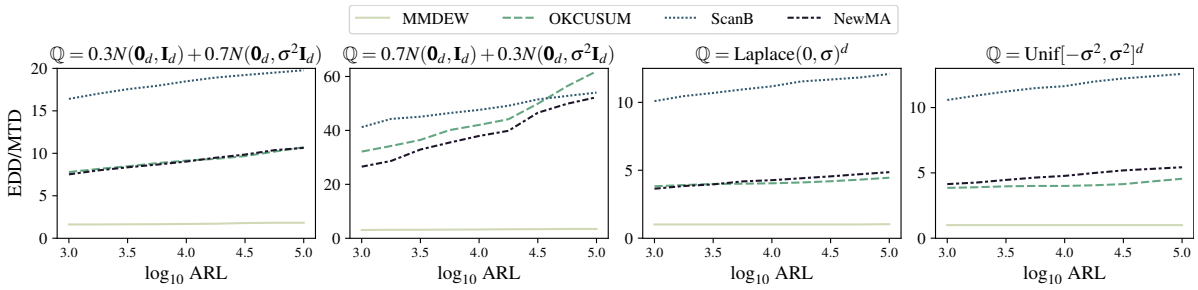

Figure 7: EDD/MTD of kernel-based change detectors with a pre-change distribution of $\mathbb{P} = \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, $d = 20$, and the indicated post-change distribution ($\sigma = 2$).

collected test statistics as threshold. For online kernel CUSUM, we set its parameters $B_{\max} = 50$ and $N = 15$, matching the settings of Wei & Xie (2022, Figure 4). Similarly, for Scan B-statistics and NEWMA, we set $B_0 = 50$; the remaining parameters of NEWMA then follow from the heuristics detailed by the authors (Keriven et al., 2020).

For approximating the EDD of MMDEW and NEWMA for a threshold b, we draw 64 and 400 samples from $\mathbb{P}$, respectively, before sampling from $\mathbb{Q}$. Online kernel CUSUM and Scan B-statistics each receive 1,000 samples from $\mathbb{P}$ upfront, for computing the variance estimate they require and to use as a reference sample. All approaches use the Gaussian kernel with the bandwidth set by the median heuristic (Garreau et al., 2018).[8]

Figure 7 collects our results, with each subfigure corresponding to a different post-change distribution, of which we sample and process 500 elements to find the first time the test statistics exceeds the threshold. We consider the mean result over 100 repetitions. The results show that OKCUSUM and NEWMA perform similarly across all experiments, with Scan B-statistics performing generally worse in three of the cases. MMDEW achieves the lowest EDD throughout. Specifically, on the mixed distribution $\mathbb{Q} = \gamma\mathcal{N}(0, \mathbf{I}_d) + (1-\gamma)\mathcal{N}(0, \sigma^2\mathbf{I}_d)$ ($\gamma = 0.3$), the EDD of the proposed method is between 1.62 and 1.82. In the more challenging setting of $\gamma = 0.7$, the EDD of MMDEW is between 3.06 and 3.46. Here, OKCUSUM performs second-best, with an EDD of 32.15 for a target ARL of 1,000 and 61.92 for a target ARL of 100,000. On the Laplace and Uniform distributions, the proposed method improves upon the results of OKCUSUM and NEWMA as well, albeit by a smaller margin.

While these experiments show that the proposed method improves upon the state-of-the-art, we note that the experiments require obtaining samples from $\mathbb{P}$, which is rarely feasible in practice. In this case, we recommend setting the threshold of MMDEW by the McDiarmid-based bound (Proposition 1), as we do in the experiments in the main text (Section 5).

D.2 Comparison with univariate approaches

In this section, we compare the test statistics of the proposed MMDEW, MMD, the Cramer-von-Mises change point model (CvM CPM; Ross & Adams 2012; Ross 2015), and the recent non-parametric Focus (Romano et al., 2023) approach on 20-dimensional multivariate normal data with a mean and correlation shift, respectively. CvM CPM and Focus handle univariate data only. To run each on multivariate data, we run one instance per dimension and consider the means of their test statistics. For MMD and MMDEW, our settings are the same as detailed in Appendix D.1. We set the pre-change distribution to $\mathbb{P} = \mathcal{N}(\mathbf{0}_d, \mathbf{I}_d)$; the respective post-change distributions $\mathbb{Q}$ are indicated in Figure 8. For CPM, which updates all previous test statistics upon observing a new sample, we report the test statistics computed after processing 500 samples from the pre-change distribution and 100 samples from the respective post-change distribution; for all other approaches, we report the test statistic computed upon observing each sample.

[8]Note that NEWMA uses random Fourier features Rahimi & Recht (2007) to approximate the kernel.